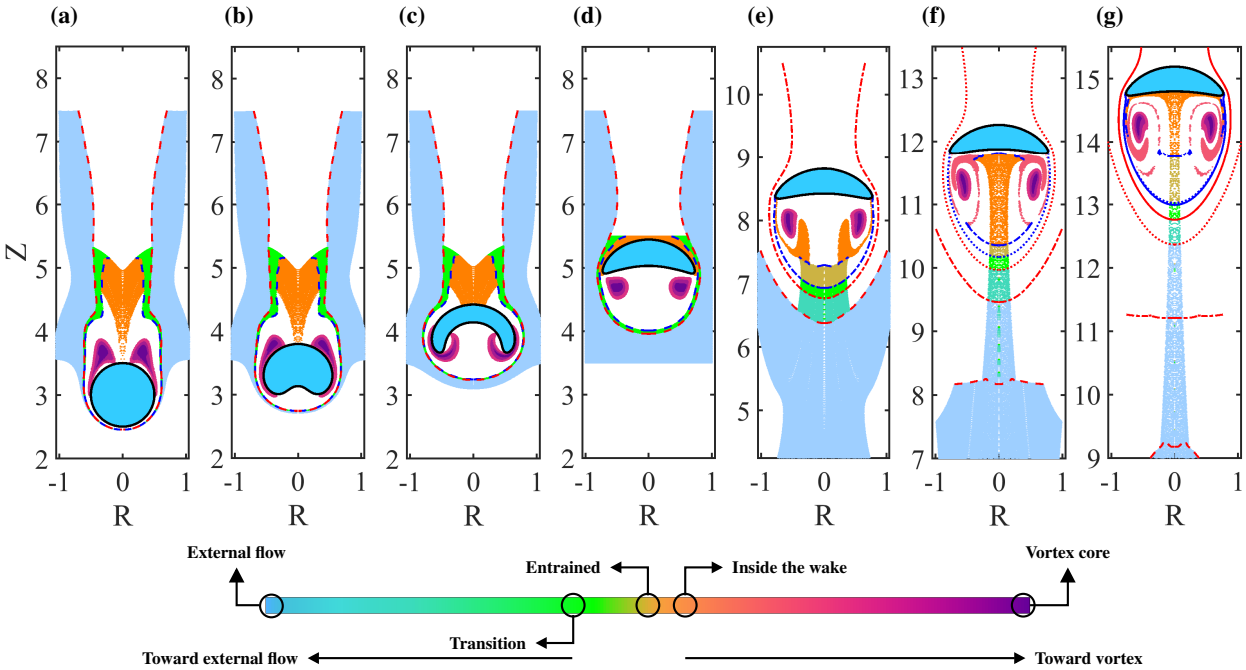

Figure 2: Temporal backward (a–c) and forward (e–g) evolution of the wake and its influence on the surrounding flow regions around the rising bubble for case C_6. The flow regions are separated by the hyperbolic LCSs obtained from the forward FTLE field with $\Delta\tau = 10$ at $\tau = 3.6$, as shown in (d), where the outermost and innermost structures are indicated by red and blue solid lines, respectively. The corresponding elliptic LCSs, shown in green with a color spectrum increasing toward the vortex core, highlight regions of rotational coherence. Panels (a–c) correspond to $\tau = 0.05$, 0.75, and 2.0, respectively, representing the backward evolution, while panels (e–g) correspond to $\tau = 5.4$, 7.0, and 8.0, showing the forward evolution. The light-blue regions represent the surrounding fluid, while the yellow and dark-blue regions denote the fluid above the bubble, with the dark-blue area located near the bubble top surface.